

Enhanced Security in Healthcare Data Exchange Using Quantum Cryptography

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Abstract— This paper introduces an innovative Secure Communication Framework tailored for Healthcare Systems, harnessing the power of Quantum Key Distribution (QKD) in tandem with Advanced Encryption Standard (AES) encryption to ensure the utmost confidentiality and integrity of Electronic Health Records (EHRs). The framework seamlessly integrates the E91 protocol, a cornerstone of secure key generation in quantum communication, thereby establishing robust quantum-secured channels among healthcare providers. Through the amalgamation of AES encryption, the framework enables comprehensive end-to-end protection for the exchange of sensitive medical data. Additionally, the implementation incorporates sophisticated security measures such as MAC ID verification and timestamp validation, further fortifying the system against potential threats. Rigorous testing against a spectrum of attacks, including Man-in-the-Middle (MITM), Denial-of-Service (DoS), and brute force attacks, underscores the framework's resilience and reliability. Despite being primarily theoretical, our approach lays a solid foundation for future exploration in quantum-secured healthcare systems, promising enhanced encryption and elevated data confidentiality.

Keywords— *Secure Communication, Healthcare Systems, Quantum Key Distribution, Advanced Encryption Standard, Electronic Health Records, E91 Protocol, MAC ID Verification, Timestamp Validation, Cybersecurity, Quantum Computing*

1. INTRODUCTION

The healthcare industry is undergoing a transformative shift towards digitalization, with electronic health records (EHRs) becoming the cornerstone of modern healthcare delivery. While the digitization of medical records offers numerous benefits such as improved accessibility, efficiency, and quality of care, it also introduces significant security challenges. Protecting the confidentiality, integrity, and availability of patient information in transit and at rest is critical to maintaining patient trust and compliance with regulatory requirements, such as the Health Insurance Portability and Accountability Act (HIPAA) in the United States.

Traditional communication methods, including email, fax, and paper-based processes, are inherently insecure and susceptible to interception, unauthorized access, and tampering. As healthcare organizations increasingly rely on electronic communication channels to exchange sensitive medical information, there is a pressing need for robust security mechanisms to mitigate the risks associated with data breaches and unauthorized disclosures.

In response to these challenges, this research proposes a Secure Communication Framework for Healthcare Systems using Advanced Encryption Standard (AES) encryption. AES is a widely adopted symmetric encryption algorithm known for its security, efficiency, and scalability. By leveraging AES encryption, the framework aims to provide end-to-end protection for medical data exchanged between healthcare providers, patients, and other authorized entities.

Additionally, to give even more security to the communication infrastructure, the framework integrates the Quantum Key Distribution (QKD) protocol E91 for secure key creation. Due to its unmatched security guarantees against key compromise attacks and eavesdropping, QKD makes it possible to generate encryption keys based on the concepts of quantum physics.

2. QUANTUM KEY DISTRIBUTION PROTOCOLS

Quantum Key Distribution (QKD) techniques rely on quantum mechanics concepts to provide secure communication. These protocols offer a unique advantage over classical cryptographic methods by providing security guarantees based on the fundamental laws of physics rather than computational complexity.

The BB84 protocol, proposed by Charles H. Bennett and Gilles Brassard in 1984, is a foundational QKD protocol that forms the basis for many subsequent developments in quantum cryptography [1]. In BB84, Alice prepares a stream of qubits (quantum bits) in one of two mutually orthogonal quantum states, namely the computational basis $\{|0\rangle, |1\rangle\}$ and the Hadamard basis $\{|+\rangle, |-\rangle\}$. She randomly selects one of these bases to encode each qubit and sends the encoded qubits to Bob. Bob independently chooses a measurement basis for each qubit and records the measurement outcomes. Following the base exchange, Alice and Bob publicly compare the bases used for encoding and measuring, discarding qubits for which they used different bases. They

then perform error correction and privacy amplification to distill a secure key from the remaining qubits. Any attempt by an eavesdropper (Eve) to intercept or measure the qubits will disturb their states, introducing errors that can be detected by Alice and Bob. The E91 protocol [2], introduced by Artur Ekert in 1991, builds upon the principles of quantum entanglement to enhance the security of key distribution.

In E91, Alice generates pairs of entangled qubits using a quantum source and sends one qubit from each pair to Bob. The qubits are typically entangled in a Bell state, such as the singlet state

$$\frac{1}{\sqrt{2}}(|01\rangle - |10\rangle)$$

Upon receiving the qubits, Bob randomly selects a measurement basis (usually the computational basis or the Hadamard basis) for each qubit and records the measurement outcomes. Following the transmission, Alice and Bob publicize their measurement bases and compare a selection of their measurement results. A secure key can be extracted by examining the correlated measurement results. The use of entangled qubits in E91 provides additional security benefits, as any attempt by an eavesdropper to intercept the qubits will disrupt their entanglement, thereby alerting Alice and Bob to the presence of an adversary.

In a comparative analysis conducted by Alvarez LC in 2016, the security strengths of the BB84 and E91 protocols were examined [3]. The BB84 protocol relies on the transmission of photons encoded in two mutually orthogonal bases, offering simplicity and efficiency in implementation. However, vulnerabilities such as susceptibility to photon number splitting and Trojan horse attacks require additional countermeasures for robust security. On the other hand, the E91 protocol leverages quantum entanglement to establish secure communication channels. This inherent resistance against intercept-resend attacks makes it a compelling choice for applications requiring high levels of security. Despite its advantages, the practical implementation of E91 presents technical challenges related to the generation and distribution of entangled qubit pairs. Therefore, while BB84 offers simplicity, E91 provides stronger security guarantees against certain types of attacks, necessitating careful consideration of the specific requirements and constraints of the application at hand.

3. LITERATURE REVIEW

The ever-increasing reliance on Electronic Health Records (EHRs) in healthcare necessitates robust security measures to protect sensitive patient information. Data breaches and unauthorized access can have devastating consequences, compromising patient privacy and hindering effective healthcare delivery. This literature review delves into existing research on securing EHR communication, focusing on various encryption techniques, access control mechanisms, and emerging technologies like quantum key distribution.

A. The paper by Pu et al. *A Lightweight and Anonymous Authentication and Key Agreement Protocol for Wireless Body Area Networks* [4] proposes a lightweight and anonymous authentication and key agreement protocol specifically designed for Wireless Body Area Networks (WBANs). WBANs require secure communication to protect sensitive health data, but resource limitations on sensor nodes necessitate lightweight protocols. This research addresses this challenge by introducing a new protocol that achieves both user anonymity (preventing identification of communicating parties) and key agreement (establishing a shared secret key for secure communication) while maintaining efficiency for resource-constrained WBAN devices. The authors evaluate the security of their protocol against various attacks commonly encountered in WBANs and demonstrate its computational efficiency compared to existing solutions.

B. *Flexible and secure access control for EHR sharing based on blockchain* by Li et al. [5] proposes a novel approach for flexible and secure access control in Electronic Health Record (EHR) sharing using blockchain technology. Blockchain offers a secure and tamper-proof platform for managing access permissions. This research introduces a new blockchain-based access control scheme that allows fine-grained control over EHR access. The system enables data owners (patients) to grant or revoke access to specific data attributes for authorized users (healthcare providers) based on pre-defined policies. This approach enhances flexibility and user control in EHR sharing while leveraging the inherent security benefits of blockchain technology for healthcare data management.

C. The literature review by McDermott et al. [6] in their paper *Electronic Health Records: A Literature Review of Cyber Threats and Security Measures* examines Electronic Health Record (EHR) systems from a cybersecurity perspective. The authors analyze existing research to identify various cyber threats that EHR systems face. These threats might include unauthorized access, data breaches, malware attacks, and insider threats. The paper also explores security measures implemented to protect

EHR data, potentially covering areas like encryption, access control, and user training. This review provides valuable insights into the cybersecurity landscape surrounding EHR systems, highlighting potential vulnerabilities and existing approaches to securing sensitive patient health information.

D. Renardi et al. focuses on securing Electronic Medical Records (EMRs) using Near Field Communication (NFC) technology in their paper *Securing Electronic Medical Record in Near Field Communication Using Advanced Encryption Standard (AES)* [7]. NFC allows for short-range wireless data exchange, potentially enabling convenient access to EMRs by authorized personnel. However, this introduces security concerns as sensitive patient data is transmitted. The research explores the use of the Advanced Encryption Standard (AES) as a potential solution for securing EMRs transmitted via NFC. AES is a widely adopted and robust encryption algorithm. By implementing AES encryption, the authors propose to safeguard EMR data during NFC communication, potentially enhancing the security of EMR access while leveraging the convenience of NFC technology.

E. Prakash et al. look into the possibilities of Quantum Key Distribution (QKD) [8] to establish secure communication channels. QKD uses quantum mechanics principles to generate a shared secret key between two parties, ensuring that only the intended receivers have access to it. The authors of the paper *Secure way of communication using Quantum Key Distribution* look into the BB84 protocol, a well-known QKD approach, and explain its implementation obstacles and benefits. They emphasize QKD's security features, particularly its resilience to eavesdropping attempts, even with the advent of powerful quantum computers. The study finishes by underlining QKD's potential to revolutionize secure communication in a variety of applications, particularly those that require sensitive data protection.

F. Rahmad et al. [9] focuses on securing Electronic Medical Records (EMRs) within the information system of a healthcare facility in their paper *Securing the electronic medical record by implementing Advanced Encryption Standard (AES) on the information system of a health service place*. The authors propose implementing the Advanced Encryption Standard (AES) as a solution to safeguard EMR confidentiality. AES is a widely recognized and robust encryption algorithm. By encrypting EMR data using AES, the research aims to protect patient information from unauthorized access within the healthcare information system. This approach offers a potential method for enhancing EMR security and ensuring patient data privacy.

4. PROPOSED METHODOLOGY

4.1. Problem Statement

The growing dependence on Electronic Health Records (EHRs) within healthcare creates a critical need for robust security solutions to safeguard sensitive patient data. Data breaches and unauthorized access pose significant threats, jeopardizing patient privacy and hindering effective healthcare delivery. Current methods for securing EHR communication often rely on pre-shared keys or traditional key exchange protocols, which can be vulnerable to compromise.

Previous research has explored various encryption methods and protocols to secure EHRs [10] [11], ranging from traditional symmetric and asymmetric encryption algorithms [12] to more advanced techniques such as homomorphic encryption and blockchain-based solutions [13]. While these approaches offer varying degrees of security, they often face challenges related to scalability, efficiency, and resilience against emerging threats. The integration of quantum key distribution with AES encryption presents a promising avenue for addressing these challenges and enhancing the security of EHR exchange in healthcare systems.

This research proposes a novel approach to address this challenge. We aim to develop a secure EHR exchange system for authorized healthcare providers that leverages the E91 protocol for provably secure key generation. This generated key will be used in conjunction with the Advanced Encryption Standard (AES) to ensure robust data encryption during communication.

4.2 E91 Protocol for Key Generation

In this section, we outline the implementation of the E91 protocol for key generation in our secure communication framework for healthcare systems. The E91 protocol leverages the phenomenon of quantum entanglement to establish a shared secret key between authorized parties. Entanglement, a uniquely quantum mechanical property, ensures that the quantum states of particles are correlated in such a way that the state of one particle is instantly determined by the state of its entangled partner, regardless of how much is the distance between them.

In our implementation, the key generation process begins with the creation of entangled qubit pairs by a quantum source. These entangled qubits are then distributed to the parties involved in the communication, namely Alice and Bob. Each party receives one qubit from each entangled pair, as shown in Fig 4.2.1 and Fig 4.2.2.

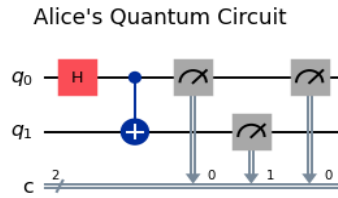


Fig 4.2.1: Alice's Quantum Circuit

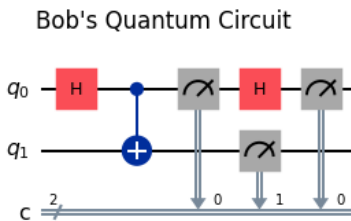


Fig 4.2.2: Bob's Quantum Circuit

The quantum operations involved in the key generation process, including qubit preparation, entanglement generation, qubit transmission, and measurement, are realized using IBM's Qiskit framework. Qiskit provides a user-friendly platform for quantum circuit design and simulation, allowing us to implement and test the E91 protocol in a simulated quantum environment. The E91 protocol offers several security features that make it highly robust against eavesdropping attacks. One of the primary advantages is that quantum states are inherently random and unpredictable, ensuring that any attempt to intercept or measure qubits will disrupt their entanglement and alert the communicating parties to the existence of an adversary. When compared to traditional key generation methods such as public-key cryptography, the E91 protocol has significant advantages in terms of security and resilience to quantum computing attacks. The E91 protocol uses quantum entanglement to provide a fundamentally safe technique for key generation that is resistant to new threats in quantum computing.

4.3 Algorithm - Secure Communication Framework Using E91 Protocol and AES Encryption

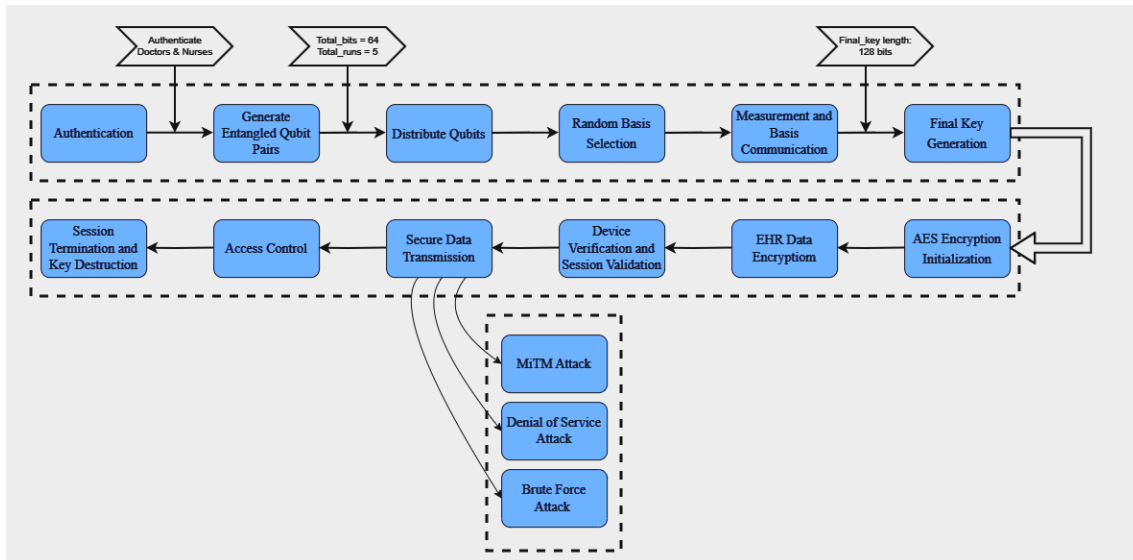


Fig 4.3.1: Block Diagram representing Secure EHR Transfer using Quantum Key Distribution and AES Encryption

This secure communication framework, as shown in Fig 4.3.1 utilizes the E91 Protocol for Quantum Key Generation, where entangled qubit pairs are created and distributed to establish a shared secret key between participants. Authentication and

random basis selection further secure the key generation process. Subsequently, the framework utilizes AES encryption with the generated key to encrypt EHR data. Device verification and session validation ensure message integrity and freshness. Finally, secure data transmission occurs, followed by access control for authorized data access. The framework also incorporates security testing mechanisms to counter potential attacks like MitM, DoS, and brute-force attempts.

i) Quantum Key Generation Using E91 Protocol

- a. *Initialization:*
 - Define **total_bits** = 64 (Number of bits in the key).
 - Define **total_runs** = 5 (Number of key generation runs).
- b. *Authentication:*
 - Alice and Bob authenticate themselves using *authenticate_party()*.
 - If authentication fails, terminate the process.
- c. *Generate Entangled Qubit Pairs:*
 - For each run in **total_runs**:
 1. Quantum Source:
 - Create entangled qubit pairs: $\{pair_i\}_{i=1}^{total_bits/2}$
 2. Distribute Qubits:
 - Distribute one qubit from each pair to Alice and the other to Bob.
- d. *Random Basis Selection:*
 - **alice_bases** = generate_random_key(total_bits).
 - **bob_bases** = generate_random_key(total_bits).
- e. *Measurement and Basis Communication:*
 - Alice measures her qubits based on **alice_bases**.
 - Bob measures his qubits based on **bob_bases**.
 - Alice and Bob exchange their bases over a classical channel.
- f. *Key Generation:*
 - For each qubit *ii*:
 - If **alice_bases[i] == bob_bases[i]**:
 - Include the measurement result in the final key.
 - **final_key** = {result_i | alice_bases[i]==bob_bases[i]}.
- g. *Repeat Steps 3-6 for total_runs and concatenate keys:*
 - **accumulated_key** == $\sum_{j=1}^{total_bits} final_key_j$

ii) Secure EHR Data Exchange Using AES

- a. *AES Encryption Initialization:*
 - Use the accumulated_key (128-bit) for AES encryption.
- b. *EHR Data Encryption:*
 - encrypted_EHR=AES_encrypt(EHR_data, accumulated_key)
- c. *Device Verification:*
 - MAC=generate_MAC(message, accumulated_key)
 - Append MAC to the encrypted message.
- d. *Session Validation:*
 - Include timestamp: timestamp=current_time()
 - Attach timestamp to the message.
 - Verify timestamp upon receipt to ensure freshness.
- e. *Secure Data Transmission:*
 - Transmit the encrypted EHR data along with MAC and timestamp.
 - Ensure only authorized parties (Alice, Bob, and nurse Charlie) can decrypt and access the data.
- f. *Access Control:*
 - Doctors Alice and Bob can modify EHR data.

- Nurse Charlie can only view EHR data.
 - Implement access control policies to enforce permissions.
- g. *Session Termination and Key Destruction:*
- After the communication window, destroy the accumulated_key.
 - Generate a new key for subsequent communication sessions.

The combination of AES encryption, MAC ID verification, and timestamp validation adds multiple layers of security to the communication process, enhancing the overall robustness of the system. By implementing these security measures, the framework ensures that sensitive medical information is safeguarded from unwanted access and tampering, maintaining patient privacy and regulatory compliance.

4.4 Security Testing

4.4.1 Man-in-the-Middle (MITM) Attack:

The system effectively defends against Man-in-the-Middle (MITM) attacks [15], a clandestine form of cyberattack where an unauthorized entity intercepts and manipulates communication between two parties. By employing AES encryption, the system ensures that data packets transferred between sender and receiver remain encrypted and indecipherable to any third-party intercepting the transmission. Mathematically, if Alice sends a message M to Bob, the encrypted message C is given by:

$$C = E_K(M)$$

where E_K is the AES encryption function using the secret key K . An interceptor (Eve) capturing C would see only the encrypted data. Without the decryption key K , Eve cannot retrieve the original message M , ensuring communication confidentiality and integrity. While AES encrypts the data, it doesn't inherently guarantee the message hasn't been tampered with during transmission. The secure data exchange protocol addresses this concern by incorporating a Message Authentication Code (MAC) alongside the encrypted data. This ensures the integrity of the message as well.

4.4.2 Denial-of-Service (DoS) Attack:

Our system implements proactive measures to mitigate Denial-of-Service (DoS) attacks [15], which seek to disrupt communication by overwhelming the system with traffic. Upon detecting an unusually high volume of requests, the system promptly initiates an automatic blocking mechanism to prevent further communication from the attacking IP address. The detection mechanism monitors the request rate R_{current} and compares it with a predefined threshold $R_{\text{threshold}}$. If $R_{\text{current}} > R_{\text{threshold}}$, where:

$$R_{\text{threshold}} = R_{\text{avg}} + \Delta R$$

With R_{avg} being the average request rate and ΔR a small tolerance, the system identifies a potential DoS attack and blocks the IP address of the source, ensuring the stability and reliability of the communication system.

4.4.3 Brute-Force Attack:

To counter Brute-Force attacks [15], our system swiftly detects and responds to unauthorized decryption attempts. When an attacker persistently tries to decrypt the encrypted messages or files by attempting different keys, the system monitors the number of unsuccessful attempts. If the number of attempts from a specific IP address IP_{attacker} exceeds a threshold $N_{\text{threshold}}$ within a defined timeframe, the system blocks the IP address. Mathematically, let K_{attacker} represent the set of keys tried by the attacker. If the number of unsuccessful attempts N from IP_{attacker} exceeds $N_{\text{threshold}}$,

$$N > N_{\text{threshold}}$$

the system triggers a blocking mechanism for IP_{attacker} , thus preventing further brute-force attempts and securing the encrypted data.

The testing results illustrate the efficacy of our secure communication system in limiting the risks posed by various cyber threats and malicious assaults. By incorporating robust encryption, authentication, and access control mechanisms, the implementation ensures the confidentiality, integrity, and availability of EHRs exchanged between healthcare providers. The resilience of the system against MITM, DoS, and Brute Force attacks underscores its suitability for safeguarding sensitive

medical information in real-world healthcare environments.

5. DISCUSSION AND FUTURE SCOPE

The proposed approach of integrating quantum key distribution with AES encryption offers several significant advantages over existing methods in securing electronic health records (EHRs) and healthcare communication systems.

5.1 Advantages

Stronger Encryption: By utilizing quantum mechanics principles for key creation, the suggested method enables the development of highly secure cryptographic keys.

Improved Data Confidentiality: The use of quantum-generated keys enhances the confidentiality of EHRs, ensuring that sensitive patient information remains protected during transmission and storage. This improved data confidentiality is essential for maintaining patient trust and compliance with regulatory requirements.

Resilience Against Security Threats: The implementation of advanced security measures, including MAC ID verification, timestamp validation, and robust encryption protocols, enhances the system's resilience against various security threats such as Man-in-the-Middle (MITM), Denial-of-Service (DoS), and brute-force attacks.

Future-Proofing Healthcare Systems: As quantum technologies evolve, combining quantum key distribution with AES encryption prepares healthcare systems to meet future security problems. This future-proofing ensures that healthcare organizations can remain ahead of emerging threats while maintaining the integrity of patient data.

5.2 Limitations

Simulation-Based Implementation: The primary limitation of the proposed approach is that it has been implemented on a simulation environment rather than on actual quantum devices. While simulations provide valuable insights and proof-of-concept demonstrations, they may not fully capture the complexities and nuances of real-world quantum systems.

Technological Maturity: Quantum technologies are still in the early stages of development, and practical implementations may face challenges related to technological maturity and reliability. Issues such as qubit stability, error correction, and environmental noise may impact the effectiveness and robustness of quantum-enabled encryption solutions.

Limited Scalability: Quantum key distribution protocols such as E91 may have limitations in terms of scalability, particularly when deployed in large-scale healthcare networks. As the number of users and devices increases, the overhead associated with key generation and distribution may become a bottleneck, affecting system performance and efficiency.

Interoperability Challenges: Integrating quantum-secured communication systems with existing healthcare IT infrastructure may pose interoperability challenges. Ensuring compatibility and seamless integration across different platforms, protocols, and standards will be essential for widespread adoption and usability.

5.3 Future Scope

Future study should concentrate on experimental validation of the suggested approach using real quantum hardware. Real-world implementations will provide significant information about the performance, scalability, and dependability of quantum-secured healthcare systems. Further research might also look into the establishment of quantum-secured communication networks within healthcare systems, which would allow for secure and efficient data interchange between healthcare practitioners, patients, and other stakeholders. Collaborative efforts to build quantum-safe encryption standards and protocols will be important in maintaining the long-term security of healthcare systems in the age of quantum computing.

Collaboration among quantum physicists, cybersecurity experts, and healthcare practitioners will be critical to the advancement and deployment of quantum-secured healthcare systems. Interdisciplinary research can tackle the difficult issues and requirements of protecting healthcare data in the quantum era.

6. CONCLUSION

In conclusion, this research introduces an innovative paradigm for ensuring secure communication within healthcare systems by amalgamating Quantum Key Distribution (QKD) with Advanced Encryption Standard (AES) encryption. By employing the E91 protocol for key generation and establishing secure communication channels, our approach presents a robust solution for safeguarding the exchange of Electronic Health Records (EHRs) among healthcare practitioners. Leveraging the tenets of quantum mechanics, our methodology ensures the generation of inherently random and safe cryptographic keys, reducing the risks associated with interception and illegal access. Furthermore, the use of AES encryption improves data security and integrity, offering a strong protection against emerging cyber threats.

Despite its theoretical character, our research highlights quantum-secured communication's transformative potential for

improving healthcare data security. Future research could look into practical implementations on quantum computing platforms, evaluating the scalability and efficacy of the proposed framework in real-world healthcare contexts. Finally, our findings contribute to the improvement of secure communication systems in healthcare, ushering in a new era of safer and more resilient healthcare data sharing.

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